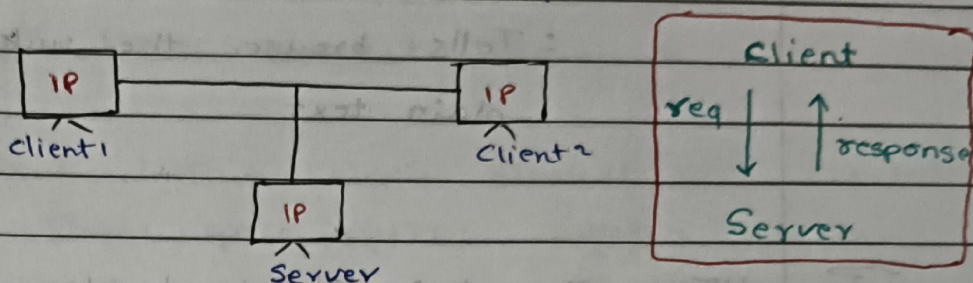




Internet :- Interconnected network of computers around the world.



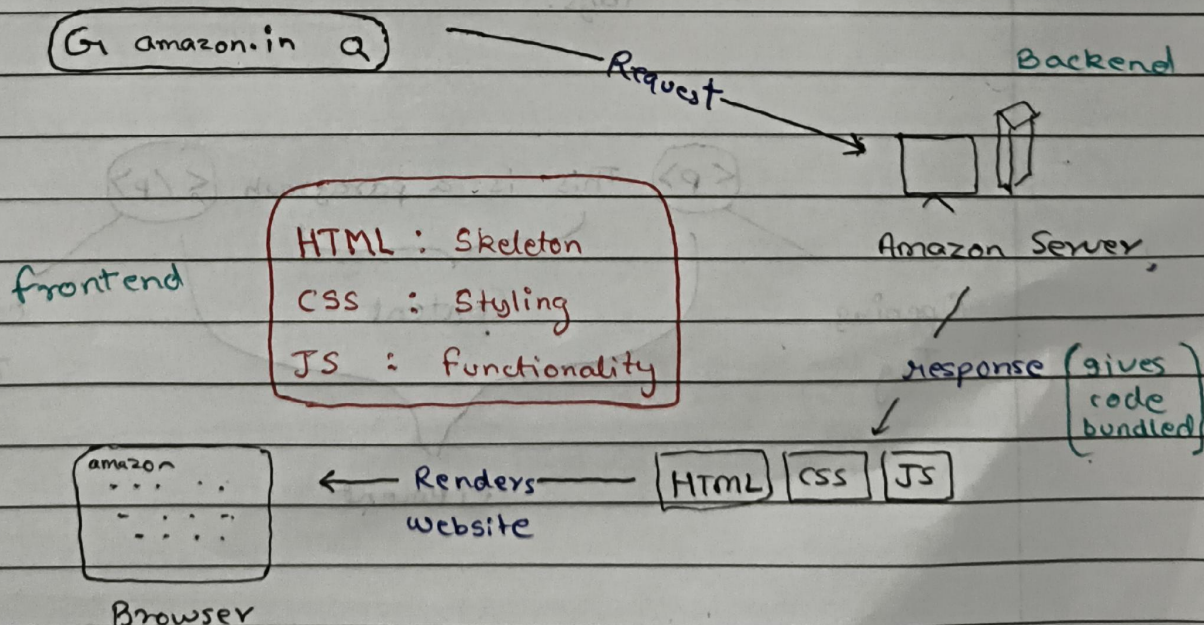
◇ Client : Someone who wants to access web/internet
Requests sends to ISP

◇ ISP : Internet Service Provider (Airtel, Jio)
looks up at DNS

◇ DNS : A dictionary for websites that contains IP addresses
Returns IP address to ISP

2/3

Web Development (Request-Response cycle)



→ HTML : HyperText Markup Language (Structure & formatting)

: Tells browser the rules how to present
• plain text

→ HTML Elements :- Standard elements that browser recognizes

Paragraph <p> </p>

Heading <h1> </h1>

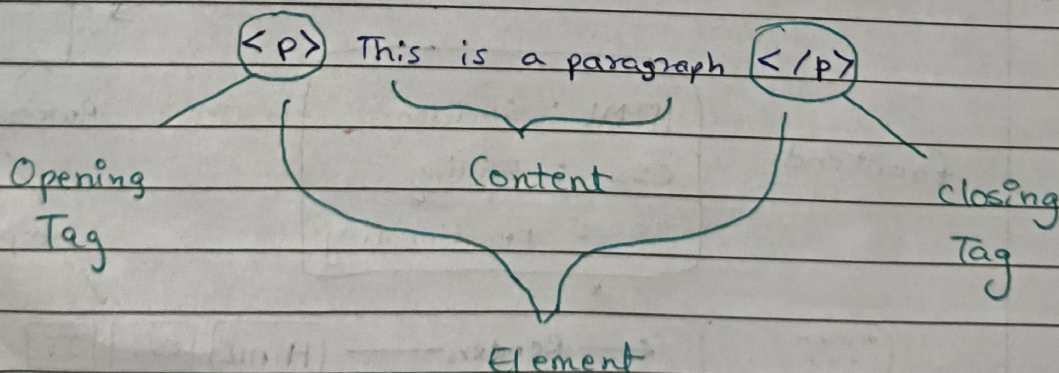
Case insensitive

interpreted same way

Image

⇒ The components used to design the structure of websites are called HTML Tags

→ HTML Tags :- A container for some content or other tags.



3 Paragraph Element

The `<p>` HTML element represents a paragraph

⚠ Content within single element are treated as single space, HTML considers only one space, no multiple spaces or paragraphs.

3 Heading Element

The `<h1>` to `<h6>` HTML elements represents six levels of section headings

➡ `<h1>` is the highest section level and `<h6>` is the lowest.

3 BoilerPlate

→ Lists in HTML

◇ Unordered list

 Bread

 Butter

 Jam

Types of ul

•	o	▪
•	o	▪
•	o	▪

To add
items to
list

◇ Ordered list

 Bread

 Butter

 Jam

Types of ol

A	a	1	i	I
B	b	2	ii	II
C	c	3	iii	III
D	d	4	iv	IV

→ HTML Attributes

Attributes are used to add more information to tag.

<html lang="en">

attribute

value

" " or ' ' are interpreted as same way in HTML.

2/3 Anchor Element : Used to add links to your page

`<a href="https://google.com">Google </a>`

attribute Complete link Text
Hyperlink reference to be written

href → attribute types

absolute links
(internet)

relative links
(local)

2/3 Image Element : Used to add image to your page

``

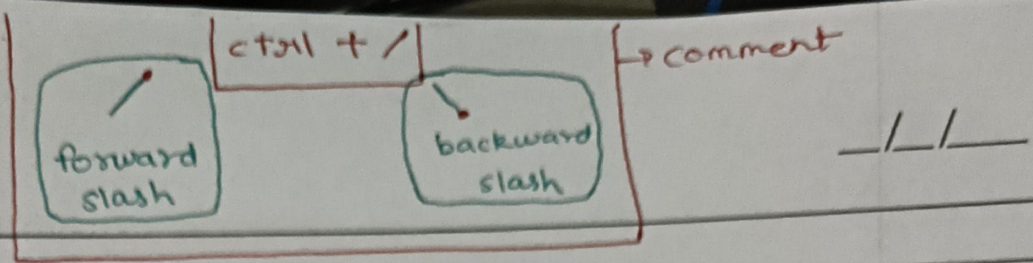
→ contains 2 attributes

→ can add extra attributes of
height = "300px" width = "300px">

→ can use absolute url in
place of img.png by pasting
image address.

→ Image tag is single
tag — no closing tag
needed.





→ More HTML tags (Pair tags : must be closed)

◆ Br Tag : Used to add next line (line break) to
`<br>` your page

◆ Bold, Italic and Underline tags : Used to highlight text
`<b>` `<i>` `<u>` in your page.

→ Comments in HTML : This is part of code that
should not be parsed

⚠ Not part of code

⚠ Ignored by browser

⚠ Used to write extra information to code

`<!-- This is HTML comment -->`

HTML is not Case Sensitive

`<html>` = `<HTML>`

`<p>` = `<P>`

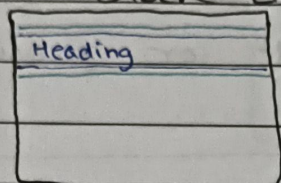
`<head>` = `<HEAD>`

⚠ Good practice to write in lower case as
in future lectures of JavaScript, it will be
case sensitive



Inline Vs Block elements

◆ Block elements : Take up full-width available
(whole block)



: start from new line

Examples : Heading elements `<h1> Hello </h1>`

Paragraph element `<p> ..... </p>`

Body element `<body> .... </body>`

◆ Inline Elements : Take up only necessary width

: Doesn't start from new line.

Examples : Anchor Tag `<a href=".....">link </a>`

Image element ``



Div element (block element) [content division element]

Div is a container used to hold other HTML elements or group elements together.

`<div>`

`<a href="www.google.com">Google </a>`

`<a href="www.youtube.com">Youtube </a>`

`</div>`

3

Span Element (Inline Element)

:- Span is also a generic container used to hold other HTML elements or group elements together.



The difference between `<div>` and `<span>` is
`<div>` is a block element whereas `<span>`
is an inline element.

:- Stays is same line.

3

Hr Tag (Single Tag)

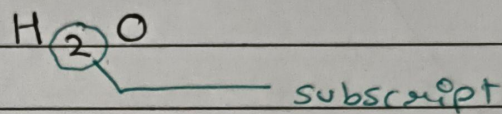
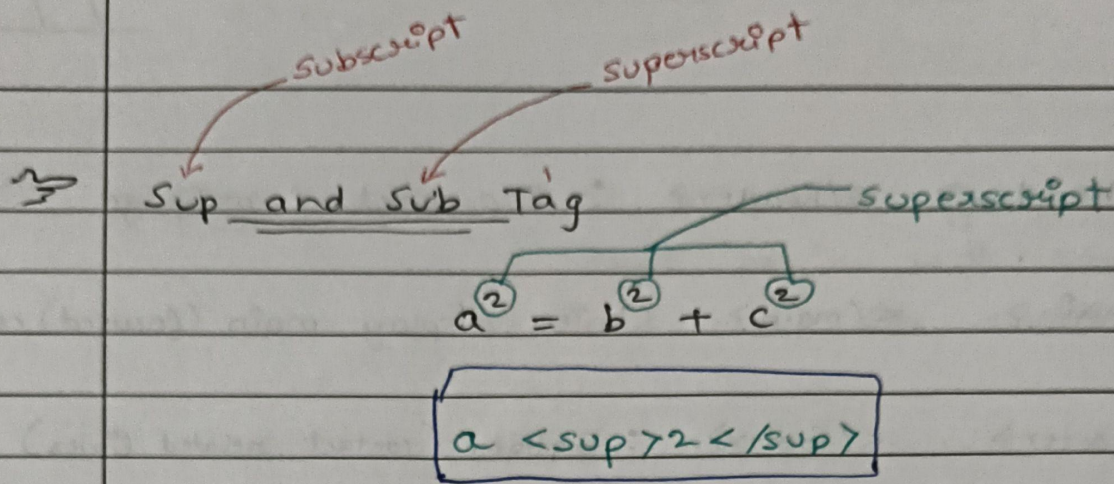
⇒ It is a horizontal rule element

⇒ Breaks page with a line



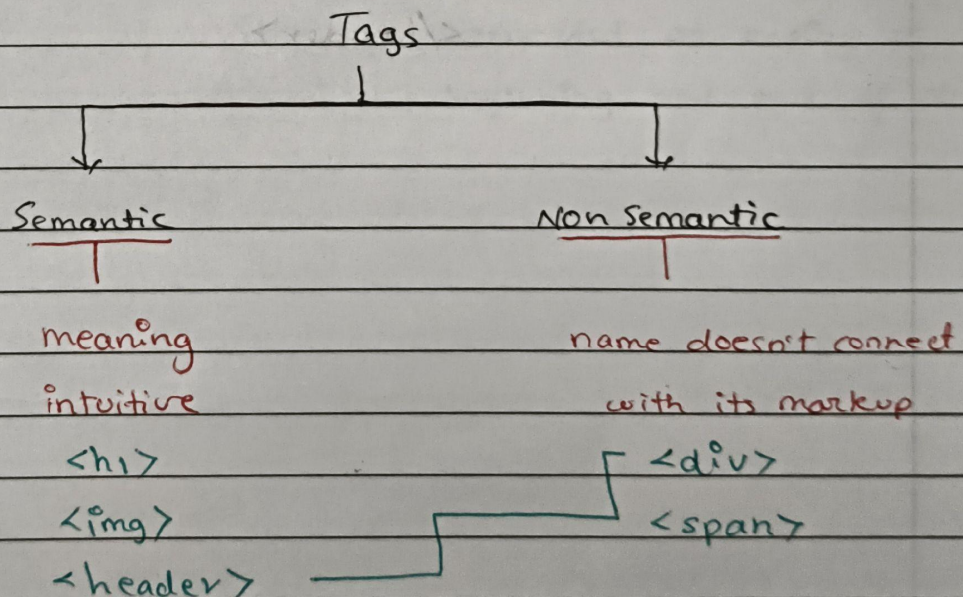
`<br>` tag is used as a line break

`<hr>` tag is used as a divider



`H <sub>2</sub> O`

→ Semantic markup : markup that relates to meaning of content.



- makes code meaningful or structured layout
- SEO friendly
- readable + screen readers (improves UX)

`<header>` `</header>` :- Top part of webpage

`<main>` `</main>` : To display main (focused) content

`<footer>` `</footer>` : Bottom (contact, related links)

`<nav>` `</nav>` : collect redirecting links.

Example : `<header>` Priyansh Kumar ~~`</header>`~~

`<nav>`

`<ul>`

`<a href="...">`

`<li>` Portfolio `</li>`

`<li>` LinkedIn `</li>`

`</ul>`

`</nav>`

`</header>`

Emmet Abbreviation

Parent-child
ul > li

multiplication
ul > li * 5

} Handy
Tools

__/__/__

2/3

HTML Entities : is a piece of text ("string") that begins with ampersand (&) and ends with semicolon (;)

: used to display reserved characters (which would otherwise be interpreted as HTML code) and invisible characters (like non-breaking spaces)

: Browser interprets them & renders correct character.

Character	Entity	Note
a & b → (&)	& amp; ;	Interpreted as beginning of entity
a < b → (<)	& lt; ;	Interpreted as beginning of a tag
a > b → (>)	& gt; ;	Interpreted as ending of a tag
a " b " → (")	& quot; ;	beginning and end of attribute value



Semantics

`<thead>` `</thead>` | `<tfoot>` `</tfoot>`
`<tbody>` `</tbody>` | `sum` `60`
~~`<tfoot>`~~

3

Tables in HTML

: Used to represent real life table data.

Name	Roll No.
Aman	01
Priyansh	21
Ansh	32

Table header `<th>`

Table Data `<td>`

Student Data

Table caption

`<table>`

`<caption> Student Data </caption>`

`<tr>`

`<th> Name </th>`

`<th> Roll No. </th>`

`</tr>`

`<tr>`

`<td> Aman </td>`

`<td> 01 </td>`

`</tr>`

`<tr>`

`<td> Priyansh </td>`

`<td> 21 </td>`

`</tr>`

`<tr>`

`<td> Ansh </td>`

`<td> 32 </td>`

`</tr>`

`</table>`

CODE

Colspan & Rowspan Attributes

: used to create cells which spans over multiple rows or columns.

<th rowspan="2">Item </th>

<th colspan="2">Price </th>

food menu

ITEM	Price	
	INR	USD
frooti	20	0.28
Chips	10	0.14

2R
2C

Forms in HTML : used to collect data from user

<form>

content...

</form>

Action attribute : used to define what action needs to be performed when a form is submitted or where the form data should be sent.

<form action="/action.php">

<form action="/action">

→ form elements

- ◇ Input : used to create multiple form controls
: there are many types of inputs that can be created using type attribute

`<input type = "text">`

password

number

time

color

button

checkbox

radio

range etc.

- ◇ Placeholder attribute (hint inside textbox)

`<input type = "text" placeholder = "Enter Name">`

- ◇ Label : represents a caption for an item in UI.
: (connects or binds)

`<label>`

Enter your username

`<input type = "text" placeholder = "Enter username">`

`</label>`

Enter your username:

★

Classic Label approach

`<label for = "username"> Enter username: </label>`

`<input type = "text" id = "username" placeholder = "...">`

id → unique name / element

→

Input Element : Button using `<input>`.

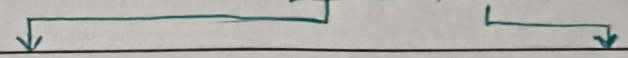
```
<input type="button">  
<input type="button" value="click me" />  
↙  
<input type="submit" />
```

⚠ Not advisable, always use `<button>`

→

Name attribute : Name of the form control.

: Submitted with the form as part of a name - value pair.



The identifier of form

The actual data entered by user

⚠ When a form is submitted, the browser sends name-value to the server.

⚠

Both `id` and `name` look similar but have purpose each

id	name
unique in whole page	not required to be unique
used for styling or scripting	used for sending data to server
Never sent to form server when submitted	Server only cares about name, not id

```
<input type="text" placeholder="enter name" id="1" name="user" />
```


➡ Input Element : checkbox (URL: age=ON)

```
<input type="checkbox" id="age" name="age">  
<label for="age"> I am 18+ </label>
```

checked

➡ Input Element : Radio Button

```
<input type="radio" name="fruit" id="apple">  
<label for="apple"> apple </label>
```

value="apple"

↓
If name="fruit" same for every button, then grouping only allows 1 value to go into URL i.e. fruit=ON

➡ Select Element : Dropdown

```
<select name="profession" id="profession">  
  <option value="student"> Student </option>  
  <option value="Teacher"> Teacher </option>  
</select>
```

selected (To pre-select)

➡ Input Element : Range

```
<label for="volume"> Volume </label>  
<input type="range" min="0" max="100" name="volume">
```

step="10"

value="70" (default)

(increase in 10 steps each time)

3

Text area

```
<label for = "feedback" > Enter feedback </label>
```

```
<textarea id = "feedback" name = "feedback" > </textarea>
```

placeholder = "...." rows = "5" cols = "100"